

TS-RAG TIME: frozen forecasting and validation mixing

1 Question and controls

Does native retrieval improve Chronos-Bolt after controlling for input context, and can a small validation interval combine it with the stronger Chronos-2 backbone? All models are frozen and evaluated as univariate median forecasts on the same official TIME windows. The controls are Chronos-T5-base at $L = 512$, Chronos-Bolt-base at $L \in \{512, 2048\}$, and Chronos-2 at $L = 8192$. Here L is the maximum number of historical observations admitted to each forecast invocation; an early window uses the available shorter history.

2 Native retrieval forecast

For one series $x_0, \dots, x_{T-1}$ and a real query origin t , let $c_t = (x_{t-512}, \dots, x_{t-1})$ be its native history. The frozen Chronos-T5-base encoder maps it to an EOS representation $e_t \in \mathbb{R}^{768}$. For every candidate origin s , the same encoder embeds c_s . Candidates may come from any item and variate in the selected dataset. Let $\tau_i(s)$ denote the calendar date of item i at index s . Eligible candidates satisfy

$$\mathcal{D}_t = \{(i, v, s) : s \geq 512, \tau_i(s + 64) \leq \tau_q(t)\}.$$

The configurable stride restricts s to $512 + j\Delta$, where $j \geq 0$ is an integer and $\Delta = 1$ by default. An optional cap retains the most recent eligible candidates. FAISS IndexFlatL2 selects $K = 10$ nearest embeddings. The released rule requests $K + 1$ candidates and discards the last returned result; fewer than 11 candidates triggers fallback.

For neighbor s_i , fetch $r_i = (c_{s_i}, x_{s_i}, \dots, x_{s_i+63}) \in \mathbb{R}^{576}$. The released ARM normalizes the complete trajectory, then encodes only its 64-point continuation with a shared MLP. Separately, Bolt’s encoder and decoder process c_t into the query hidden representation. Multihead attention and learned gates fuse that decoder representation with the K neighbor representations. A residual update and Bolt’s output projection produce 64 points at each quantile; retain the median. Fusion is after the query decoder, not before it.

For task horizon $H \leq 64$, crop one native forecast. For $H > 64$, append each median chunk, retain the last 512 values, re-embed and retrieve, and repeat. The observed candidate cutoff remains t throughout recursion: predicted future values never grant access to real observations after t .

3 One-axis retrieval ablation

The retired 2^4 factorial schedule was too long and never completed its first new cell. Its replacement keeps native $L = 512$, $K = 10$, the 64-point continuation and every released default fixed while changing one setting at a time:

Label	Only change
<code>tsrag_same_series</code>	query item/variate only
<code>tsrag_aligned</code>	query calendar phase required
<code>tsrag_inst_l2</code>	IN-lookback L2 replaces T5 EOS
<code>tsrag_query_scale</code>	neighbor aligned to query scale

Released `tsrag` is the shared baseline. Axis interactions are not tested. Date-phase periods count observations and follow Adapttime’s dataset protocol.

For neighbor lookback mean/std (μ_i, σ_i) and query statistics (μ_q, σ_q) , with standard deviations floored at 10^{-8} , align the full neighbor trajectory by

$$r'_i = \frac{r_i - \mu_i}{\sigma_i} \sigma_q + \mu_q.$$

Only the 512-point lookback enters neighbor statistics. T5 selection embeds the aligned lookback at each query; fusion uses the query’s Bolt normalization rather than re-estimating full-neighbor statistics, which would cancel alignment.

For the alternative distance, cache lookback instance normalization:

$$z(c) = \frac{c - \mu(c)}{\max\{\text{std}(c), 10^{-8}\}}.$$

Rank by squared L2 over finite overlapping coordinates, multiplying by $512/m$ for m observed overlaps and requiring at least $\lceil 0.8 \cdot 512 \rceil$. This selection uses no T5. Affine query-scale alignment preserves IN L2 rankings, while the query-scale variant still changes fusion.

4 Fallback and validation mixture

The TS-RAG forecast R_t is replaced as one complete window by the independent Bolt-max ($L = 2048$) median B_t for insufficient history/candidates, extraction or model errors, or non-finite native predictions. Preserve masks, counts and reasons; the fallback is part of the implemented method. Comparison reports aggregate fallback counts, rates and causes on test alone; validation fallback artifacts remain separate diagnostics for the frozen mixture.

Let C_t denote the maximum-context Chronos-2 median. Validation dates lie strictly before the official test interval. For variate v , define

$$q_{tv} = \max\{\text{std}(c_{tv}), 10^{-8}\}, \quad \ell_{tv}(F) = \frac{1}{|A_{tv}|} \sum_{h \in A_{tv}} \left(\frac{F_{tvh} - x_{t+h,v}}{q_{tv}} \right)^2,$$

where A_{tv} contains finite observed target steps. Admit only finite paired scores. Average ℓ_{tv} over supported variates at each series/date. For n usable dates, let w count dates where TS-RAG has lower averaged loss, with a tie contributing one half. The Beta(1, 1)-smoothed weight is

$$p = \frac{1 + w}{2 + n}, \quad M_t = (1 - p)C_t + pR_t.$$

Freeze p before testing. With $n = 0$, explicitly use $p = 0$ and pure Chronos-2. This is a smoothed win frequency, not a fitted optimal ensemble weight.

5 Evaluation and attribution

The shared Seasonal Naive artifact fixes finite target support and eligible metric cells. Every ablation variant uses that same grid and TIME’s median metrics. Task summaries report mean, population variance and standard deviation over finite series-window-variate cells; this dispersion is not across-run seed uncertainty. Reports retain coverage, fallback rates and pipeline timings. No benefit is claimed until new cluster results are obtained and analyzed.

6 Borrowed work

ARM and retrieval are source-adapted from [TS-RAG revision 73ac807](#). The local Bolt forecasting core is source-adapted from [Chronos revision 7dc4435](#). Lookback IN, overlap L2, query-scale alignment and date periods are narrowed from Adapttime revision 33e75400. Calendar-causal datastore construction, rollout, fallback, one-axis orchestration and validation mixing are project-specific. Training and alternate ARM augmentation paths are omitted.